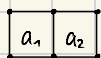



Two-site toy model



2-site lattice representing the 2 possible states.

$$\text{initial state } |\psi(0)\rangle = |a_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|a_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

No Potential

String will move from one state to another with an amplitude J :

$$\text{Hamiltonian: } H = \begin{pmatrix} 0 & -J \\ -J & 0 \end{pmatrix}$$

$$\text{eigenvalues: } E_{\pm} = \pm J \quad \begin{cases} -J: \text{ground state} \\ +J: \text{excited state} \end{cases}$$

$$\text{eigenvalues: } E = -J \rightarrow |\varphi_1\rangle = \frac{1}{\sqrt{2}} (|a_1\rangle + |a_2\rangle) \quad \text{superposition of}$$

$$E = +J \rightarrow |\varphi_2\rangle = \frac{1}{\sqrt{2}} (|a_1\rangle - |a_2\rangle) \quad \text{possible states.}$$

$$\text{initial state: } |\psi(0)\rangle = \frac{1}{\sqrt{2}} (|\varphi_1\rangle + |\varphi_2\rangle)$$

$$|\psi(0)\rangle = |a_1\rangle$$

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} (|a_1\rangle + |a_2\rangle + |a_1\rangle - |a_2\rangle) = \frac{1}{\sqrt{2}} (|a_1\rangle + |a_1\rangle)$$

normalized initial state!

$$\text{time evolution: } |\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{iJt} |\varphi_1\rangle + e^{-iJt} |\varphi_2\rangle \right)$$

$$= \frac{1}{\sqrt{2}} \left[e^{iJt} \frac{1}{\sqrt{2}} (|a_1\rangle + |a_2\rangle) + e^{-iJt} \frac{1}{\sqrt{2}} (|a_1\rangle - |a_2\rangle) \right]$$

$$= \frac{1}{2} \left[e^{iJt} |a_1\rangle + e^{iJt} |a_2\rangle + e^{-iJt} |a_1\rangle - e^{-iJt} |a_2\rangle \right]$$

$$= \frac{e^{iJt} + e^{-iJt}}{2} |a_1\rangle + \frac{e^{iJt} - e^{-iJt}}{2} |a_2\rangle$$

$$= \cos(Jt) |a_1\rangle + \sin(Jt) |a_2\rangle$$

$$\text{Probability to find the string in } |a_2\rangle: P(|a_2\rangle) = |\langle a_2 | \psi(t) \rangle|^2 = |\sin(Jt)|^2 = \sin^2(Jt)$$

→ Oscillation between states with frequency given by energy J .

observable of $\hat{H}$

$$\text{To calculate energy expectation value: } \langle \psi(0) | \hat{H} | \psi(0) \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & -J \\ -J & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

→ Makes sense because the initial state is a superposition of states with $E = \pm J$.

Energy is fully kinetic because no potential.

Adding a Potential

Add potential V to $|a_1\rangle \rightarrow H = \begin{pmatrix} V & -J \\ -J & 0 \end{pmatrix}$

$$\det(H - E \mathbb{1}) = \begin{vmatrix} V-E & -J \\ -J & -E \end{vmatrix} = -E(V-E) - J^2 = -EV + E^2 - J^2 = 0$$

$$E^2 - EV - J^2 = 0$$

eigenvalues $\left\{ \begin{array}{l} E = \frac{V}{2} \pm \sqrt{\frac{V^2}{4} + J^2} \\ E_- = \frac{V}{2} - \sqrt{\frac{V^2}{4} + J^2} \\ E_+ = \frac{V}{2} + \sqrt{\frac{V^2}{4} + J^2} \end{array} \right.$

How to see $P(a_2)$ is small without calculating it?

Eigenstates

$$(H - E_i \mathbb{1}) |\phi_i\rangle = 0, \quad |\phi_i\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\begin{pmatrix} V-E_i & -J \\ -J & -E_i \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0$$

$$\left\{ \begin{array}{l} (V-E_i)c_1 - Jc_2 = 0 \\ -Jc_1 - E_i c_2 = 0 \end{array} \right\} \quad c_2 = -\frac{Jc_1}{E_i} \rightarrow |\phi_i\rangle = N \begin{pmatrix} 1 \\ -J/E_i \end{pmatrix}$$

$$\langle \phi_i | \phi_i \rangle = 1 \rightarrow \left(N^2 + N^2 \frac{J^2}{E_i^2} \right) = 1$$

$$N = \frac{1}{\sqrt{1 + J^2/E_i^2}}$$

$V \gg J$:

$$E_{\pm} = \frac{V}{2} \pm \sqrt{\frac{V^2}{4} + J^2} = \frac{V}{2} \pm \frac{V}{2} \sqrt{1 + \frac{4J^2}{V^2}} \approx \frac{V}{2} \pm \frac{V}{2} \left(1 + \frac{2J^2}{V^2} \right) \rightarrow E_{-} \approx \frac{V}{2} - \frac{V}{2} - \frac{J^2}{V} = -\frac{J^2}{V}$$

Taylor $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$

$$E_{+} \approx \frac{V}{2} + \frac{V}{2} + \frac{J^2}{V} = V + \frac{J^2}{V}$$

giving eigenvectors $|\phi_{-}\rangle \approx \frac{1}{\sqrt{1 - J^2/V^2}} \begin{pmatrix} 1 \\ V/J \end{pmatrix} = \frac{1}{\sqrt{1 - V^2/J^2}} \begin{pmatrix} 1 \\ V \end{pmatrix} \rightarrow \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |a_2\rangle$

$$\frac{1}{\sqrt{1 - \frac{V^2}{J^2}}} \rightarrow 0 \text{ as } V \rightarrow \infty$$

$$\frac{V}{\sqrt{1 - \frac{V^2}{J^2}}} \rightarrow \frac{V}{V} = 1 \text{ as } V \rightarrow \infty$$

$$|\phi_{+}\rangle \approx \frac{1}{\sqrt{1 - \frac{J^2}{(V + \frac{J^2}{V})^2}}} \begin{pmatrix} 1 \\ -J/V + \frac{J^2}{V} \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |a_1\rangle$$

$$\frac{1}{\sqrt{1 - \frac{J^2}{V^2}}}$$

$$\frac{-J}{V + \frac{J^2}{V}} \cdot \frac{1}{\sqrt{1 - \frac{J^2}{V^2}}} \rightarrow \frac{-J}{V} \rightarrow 0$$

→ The eigenstates are localized around the 2 states now, instead of around the + and - superposition of the states when no potential.

Using 2-level Rabi model: $H = \begin{pmatrix} \Delta & -J \\ -J & \Delta \end{pmatrix}$, $\Delta = \frac{V}{2} \leftarrow$ shifting zero-point of energy.

Eigenvalues: $E_{\pm} = \pm \sqrt{J^2 + \Delta^2}$

Eigenstates: $| \sigma_+ \rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$, $| \sigma_- \rangle = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$, $\tan(2\theta) = \frac{J}{\Delta}$ Derivation ex. 1.11

$\hookrightarrow | \alpha_1 \rangle = \cos(\theta) | \sigma_+ \rangle - \sin(\theta) | \sigma_- \rangle$

$| \alpha_2 \rangle = \sin(\theta) | \sigma_+ \rangle + \cos(\theta) | \sigma_- \rangle$

Evolve state in time: $| \psi(t) \rangle = \cos(\theta) e^{-iE_+t} | \sigma_+ \rangle - \sin(\theta) e^{-iE_-t} | \sigma_- \rangle$

$\langle \alpha_2 | \psi(t) \rangle = \cos(\theta) e^{-iE_+t} \sin(\theta) - \sin(\theta) e^{-iE_-t} \cos(\theta)$
 $= \sin(\theta) \cos(\theta) [e^{-iE_+t} - e^{-iE_-t}]$

$P(\alpha_2)(t) = |\langle \alpha_2 | \psi(t) \rangle|^2 = \sin^2(\theta) \cos^2(\theta) \underbrace{\left| e^{-iE_+t} - e^{-iE_-t} \right|^2}_{4 \sin^2\left(\frac{(E_+ - E_-)t}{2}\right)}$

$E_{\pm} = \pm \sqrt{J^2 + \Delta^2} \rightarrow P(\alpha_2)(t) = (\sin(\theta) \cos(\theta))^2 4 \sin^2\left(\frac{2 \sqrt{J^2 + \Delta^2}}{2} \cdot t\right)$
 $= \left(\frac{\sin(2\theta)}{2}\right)^2 4 \sin^2\left(\sqrt{J^2 + \Delta^2} \cdot t\right) = \sin^2(2\theta) \sin^2(\sqrt{J^2 + \Delta^2} \cdot t)$

$\tan(2\theta) = \frac{J}{\Delta} \rightarrow \sin(2\theta) = \frac{\tan(2\theta)}{\sqrt{1 + \tan^2(2\theta)}} = \frac{J}{\sqrt{J^2 + \Delta^2}}$ Matte 4 Formelblatt

Expectation value: $\langle T \rangle(t) = \frac{-J^2}{\sqrt{J^2 + \Delta^2}} \cdot \sin\left(2 \sqrt{J^2 + \Delta^2} \cdot t\right)$

$\rightarrow$ Transition heavily suppressed when large potential (many strings)